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Microaeration for hydrogen sulfide removal in UASB reactor

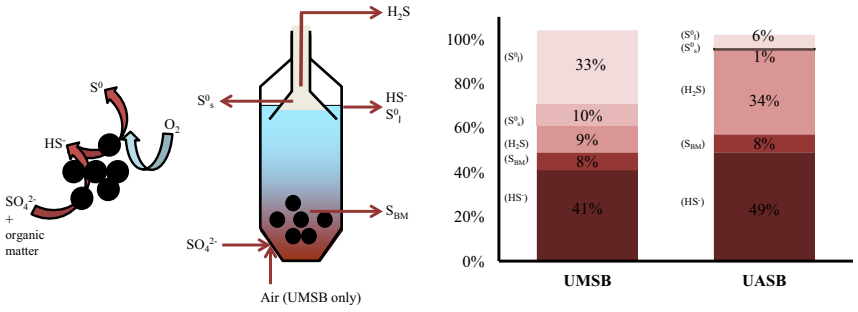
Lucie Krayzelova [a,*], Jan Bartacek [a], Nina Kolesarova [b], Pavel Jenicek [a]

[a] Department of Water Technology and Environmental Engineering, Institute of Chemical Technology Prague, Technicka 5, 166 28 Prague 6, Czech Republic
[b] Department of Chemical and Environmental Engineering, Slovak University of Technology in Bratislava, Radlinskeho 9, 812 37 Bratislava, Slovakia

HIGHLIGHTS

- Innovative application of microaeration directly in UASB reactor.
- Reliable and stable H_2S removal efficiency in long-term operation.
- Sulfur and COD balance in microaerated UASB reactor.
- Evaluation of the effects of microaeration on the activity of granular sludge.

GRAPHICAL ABSTRACT



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ABSTRACT

The removal of hydrogen sulfide from biogas by microaeration was studied in Up-flow Anaerobic Sludge Blanket (UASB) reactors treating synthetic brewery wastewater. A fully anaerobic UASB reactor served as a control while air was dosed into a microaerobic UASB reactor (UMSB). After a year of operation, sulfur balance was described in both reactors. In UASB, sulfur was mainly presented in the effluent as sulfide (49%) and in biogas as hydrogen sulfide (34%). In UMSB, 74% of sulfur was detected in the effluent (41% being sulfide and 33% being elemental sulfur), 10% accumulated in headspace as elemental sulfur and 9% escaped in biogas as hydrogen sulfide. The efficiency of hydrogen sulfide removal in UMSB was on average 73%. Microaeration did not cause any decrease in COD removal or methanogenic activity in UMSB and the elemental sulfur produced by microaeration did not accumulate in granular sludge.

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1. Introduction

Anaerobic treatment of wastewaters from industries such as pharmaceutical, paper, leather or food processing with high amount of sulfate in the influent often leads to high amount of hydrogen sulfide in biogas and sulfide (HS^- and S^{2-}) in liquid phase (Sawyer et al., 2003; Sinbuathong et al., 2007):

$$SO_4^{2-} + \text{organic matter} \xrightarrow{\text{anaerobic bacteria}} HS^- + H_2O + HCO_3^- \quad (1)$$

$$S^{2-} \overset{H^+}{\leftrightarrow} HS^- \overset{H^+}{\leftrightarrow} H_2S \quad (2)$$

High concentrations of hydrogen sulfide in biogas cause the corrosion of concrete and steel and emissions of odor and sulfur dioxide during the combustion of biogas (Buisman et al., 1990; Hulshoff Pol et al., 1998). Moreover, sulfide in liquid phase is toxic to methanogens and causes the inhibition of anaerobic process (Khanal

Abbreviations: COD, chemical oxygen demand; CSTR, completely stirred tank reactor; DO, dissolved oxygen; G–L–S separator, gas liquid solid separator; HRT, hydraulic retention time; SMA, specific methanogenic activity; TSS, total suspended solids; UASB, Up-flow Anaerobic Sludge Blanket; UMSB, Up-flow Microaerobic Sludge Blanket; VSS, volatile suspended solids.

* Corresponding author. Tel.: +420 220 443 143.
E-mail address: lucie.krayzelova@vscht.cz (L. Krayzelova).

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and Huang, 2003; Zhou et al., 2007). For these reasons, the removal of sulfide from both gaseous and liquid phase is required and beneficial. Hydrogen sulfide can be removed from contaminated streams (liquid or gaseous) by physico-chemical processes such as adsorption, precipitation or reaction with alkaline substances (Abatzoglou and Boivin, 2009). Physico-chemical methods for hydrogen sulfide removal from biogas need additional chemicals and they work under high temperatures and pressures. This makes these methods energetically demanding and expensive (Appels et al., 2008). In contrast, biological methods have lower operational costs with lower or none utilization of chemicals (Syed et al., 2006). Biological methods are most often based on the biochemical oxidation of sulfide to sulfate, thiosulfate or elemental sulfur (Díaz et al., 2011).

Microaeration (i.e. dosing of small amount of air or oxygen into anaerobic reactor) causes the oxidation of sulfide to elemental sulfur according to the following equation (Janssen et al., 1995):

$$2HS^- + O_2 \rightarrow 2S^0 + 2OH^- \quad (3)$$

It is an efficient and stable method for hydrogen sulfide removal in anaerobic digesters and biogas plants (Botheju and Bakke, 2011; Díaz and Fdz-Polanco, 2012; Jenicek et al., 2008, 2010; Ramos et al., 2013; Zhu et al., 2009). Unlike other biological methods, microaeration does not require separate reactor or complex control system (Janssen et al., 1995). So far, microaeration has been only few times reported in reactors for wastewater treatment such as Up-flow Anaerobic Sludge Blanket (UASB) reactor (Zheng and Cui, 2012; Zhou et al., 2007). Moreover, the focus of these studies was not the removal of hydrogen sulfide from biogas, but alleviation of sulfide toxicity (Zhou et al., 2007) or simultaneous COD removal and nitrification (Zheng and Cui, 2012).

In this work, the effect of microaeration in UASB reactor treating synthetic brewery wastewater was investigated and compared with a non-aerated UASB reactor. The fate of sulfur in both reactors including sulfur and COD balance was described. Moreover, the influence of microaeration on the methanogenic activity of the granular sludge present in the microaerobic reactor was evaluated.

2. Methods

2.1. Experimental set-up

Two reactors were operated in this study (Fig. 1): Up-flow Anaerobic Sludge Blanket reactor (UASB) and Up-flow Microaerobic Sludge Blanket reactor (UMSB) as shown previously by Zheng et al. (2013). Both reactors (working volume 2.7 L) were kept at 37 °C with temperature controlled by resistance wire wrapped around the reactor. Reactors were inoculated with 1 L of granular sludge (particle size up to 0.5 cm) from a full scale UASB reactor treating wastewater from candy production. The average concentration of total suspended solids (TSS) was 50 g L^{-1} of reactor. Both reactors were fed with synthetic brewery wastewater at loading rate 8 g COD L^{-1} d^{-1}, and sulfate loading rate 0.084 g S L^{-1} d^{-1} (COD/S ratio of 95). The dosage of sulfate was artificially elevated compared to typical brewery wastewater in order to achieve high concentration of hydrogen sulfide in biogas. Hydraulic retention time (HRT) was 8 h, recycle flow 8 L h^{-1} and superficial up-flow velocity 1.2 m h^{-1}. The operation of both, UMSB and UASB reactor began at the same time under the same anaerobic conditions. Air was dosed with a peristaltic pump into the recycled effluent of the UMSB reactor. Air dosing started after a start-up period (2.5 months) and the amount of air was 1 L d^{-1}.

This amount corresponded to the stoichiometric molar ratio of O_2/S assuming that sulfate from the influent was first reduced to sulfide and subsequently oxidized to elemental sulfur (see Eqs. (1) and (3)). UASB reactor was kept under anaerobic conditions and was used as a control. Both UMSB and UASB reactors were operated for 373 days.

2.2. Synthetic brewery wastewater

Synthetic brewery wastewater used in this study was prepared based on Peng et al. (2006) and Tarn (2002) with an extra addition of sulfate. It contained (per liter of influent): malt extract 5 g; yeast extract 2.5 g; peptone 0.75 g; maltose 4.3 g; 95% ethanol 6 ml; NH_4Cl 1.18 g; KH_2PO_4 0.26 g; $MgSO_47H_2O$ 2.6 g; $CaCl_22H_2O$ 0.06 g and $NaHCO_3$ 5.6 g. This concentrated solution was diluted (1:5) by tap water before injection into the reactors. Trace elements solution was not added into the influent as all micronutrients were expected to be present in the malt extract. The average pH of the concentrated solution was, thanks to the creation of volatile fatty acids, between 5.7 and 5.8. Due to the sufficient buffering capacity of the dilution water, pH inside of the reactor was always between 7.0 and 7.6.

2.3. The assessment of specific methanogenic activity

The assessment of specific methanogenic activity (SMA) was carried out at mesophilic conditions in serum bottles (total volume of 120 mL) without mixing in triplicates according to the guidelines proposed by Angelidaki et al. (2009). The volume of liquid and gas phase was 80 and 40 mL, respectively. The synthetic brew-

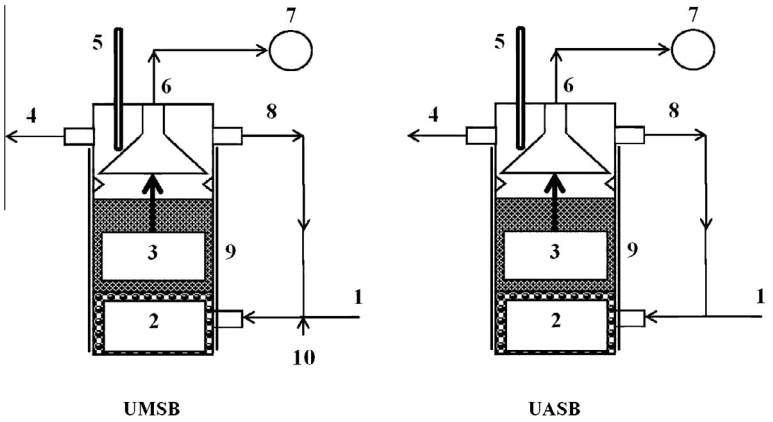


Fig. 1. Scheme of UASB and UMSB reactor used in this study. (1) Influent, (2) distribution bed, (3) granular sludge, (4) effluent, (5) thermometer, (6) biogas, (7) gas-meter, (8) recycle, (9) temperature control, (10) air dosing.

ery wastewater was used as a substrate with organic loading 2 g of substrate per g of volatile suspended solids (VSS) of inoculum. One gram (wet weight) of granular sludge was used as inoculum. The production of biogas was measured volumetrically using water displacement method (Angelidaki et al., 2009). The biogas composition was determined by GC 8000^{TOP} (Fisons Instruments, USA).

In order to estimate potential trace metal limitation in the granular sludge, targeted SMA experiments with extra addition of trace metal mixture according to Fermoso et al. (2010) were carried out occasionally.

2.4. Analytical methods

Influents and effluents of both reactors were sampled for COD_{Cr}, pH, VSS, TSS and sulfide analysis. Granular sludge was sampled and analyzed for TSS, VSS and elemental composition. Analysis of COD_{Cr}, VSS and TSS were done according to the Standard Methods (American Public Health Association, 1997) as well as the determination of total sulfide, dissolved sulfide and hydrogen sulfide by iodometric titration. The amount of methane and carbon dioxide in biogas was measured by GC 8000^{TOP} (Fisons Instruments, USA) equipped with a heat conductivity detector HWD 800. The elemental composition of sludge was assessed by Elemental Vario EL III (Elementar Analysensystem GmbH, Germany) and by X-ray fluorescence analysis using the ARL 9400 XP sequential WD-XRF spectrometer (THERMO ARL, Switzerland).

3. Results and discussion

3.1. The efficiency of hydrogen sulfide removal from biogas by microaeration

The concentration of hydrogen sulfide in biogas in UMSB reactor decreased from over 9.0 g m^{-3} to 1.0 g m^{-3} in less than 3 days upon the start of aeration (Fig. 2). After 10 days, the concentration of hydrogen sulfide in the biogas produced in UMSB reactor decreased to almost zero and remained at that level for almost a month. After 30 days, the concentration of hydrogen sulfide in biogas in UMSB occasionally increased up to 6.0 g m^{-3}, but it remained significantly lower compared to UASB reactor for 299 days of operation. These peaks of hydrogen sulfide concentration can be attributed to the uneven actual biogas production which caused temporary oversaturation of the effluent by sulfide. This problem is intrinsic to the small-scale set-up and should be alleviated by the future scale-up of the reactor. The average concentration of hydrogen sulfide in biogas of UASB reactor was 8.9 ± 1.4 g m^{-3} while in UMSB reactor it was only 2.2 ± 1.3 g m^{-3}.

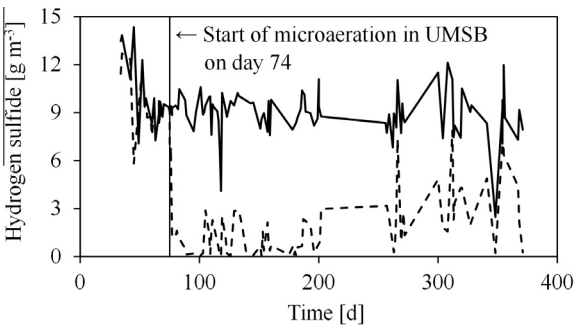


Fig. 2. The concentration of hydrogen sulfide in biogas from UMSB (---) and UASB (—) reactor throughout the operation.

The dose of air corresponding to the stoichiometric ratio of O_2/S (0.5 mol O_2 per 1 mol of S^{2-}) needed for oxidation of sulfide to elemental sulfur removed 73% of hydrogen sulfide from the biogas of UMSB reactor (with fluctuations between 20% and 99%).

No negative effect on the methanogenic activity of the biomass was observed. The efficiency of desulfurization was not as high as in other studies (Díaz et al., 2011; Jenicek et al., 2010), where the removal efficiency of 98–99% was observed. However, the dose of air or oxygen was much higher in those studies compare to present study. Further research will be focused on the optimization and control of the amount of air in order to ensure complete desulfurization without oxygen being present in biogas.

3.2. Sulfur balance

Microaeration had significant and fast effect on sulfur distribution in UMSB reactor (Fig. 3). Five major streams of sulfur leaving UMSB and UASB were identified: (1) total sulfide in the effluent, (2) excess granular sludge (biomass growth), (3) sulfur in biogas (as hydrogen sulfide), (4) deposition in the headspace of the reactors (mainly gas–liquid–solid, G–L–S separator), and (5) solids in the effluent (excluding sulfide). The main difference between UASB and UMSB reactors were in the amount of hydrogen sulfide in biogas, in the deposition of sulfur in the G–L–S separator, and solids in the effluent excluding sulfide.

Of all sulfur dosed into UASB reactor, 34% was found in biogas while only 9% of sulfur was detected in biogas from UMSB. The sulfide detected in the effluent from UMSB and UASB reactors accounted for 41% and 49% of the sulfur dosed, respectively. I.e. the addition of air into UMSB reactor decreased not only the concentration of hydrogen sulfide in biogas, but also the concentration of sulfide in the effluent. The average effluent concentration from UASB and UMSB was 13.6 mg L^{-1} and 11.5 mg L^{-1}, respectively. These results were consistent with Díaz et al. (2011) who found that, while microaeration with air dosed into the gas phase removed only hydrogen sulfide from biogas, dissolved sulfide was removed if air was partially dosed into the liquid phase. Only negligible amounts of sulfate and thiosulfate were found in the effluents. This could be explained by the very low concentration of dissolved oxygen (DO). Annachhatre and Suktrakoolvait (2001) observed that at DO concentration greater than 0.1 mg L^{-1}, sulfate was the main end product, while at DO concentration less than 0.1 mg L^{-1}, elemental sulfur was the main end product. DO concentrations above 0.1 mg L^{-1} have never been observed in this study. Krishnakumar et al. (2005) controlled the treatment of sulfide and the creation of different sulfur species based on ORP. At ORP from −400 to −350 mV, sulfide removal was nearly 100% with

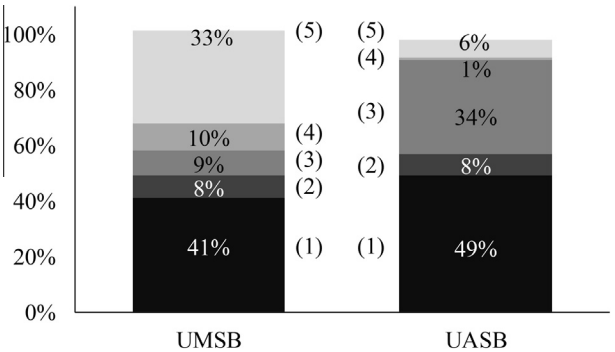


Fig. 3. Sulfur balance calculated in UMSB and UASB reactor during days 74–373 of operation: (1) total sulfide in the effluent; (2) sulfur in excess sludge; (3) hydrogen sulfide in biogas; (4) elemental sulfur in G–L–S separator; and (5) solids in the effluent (excluding sulfide).

the sulfur recovery around 80% and sulfate around 2–3%. ORP has never exceeded −420 mV in UMSB reactor. The average ORP was −450 mV and −425 mV for UASB and UMSB reactor, respectively.

Significant difference was also observed in sulfur detected in the effluent. Over 33% of sulfur was present in the solid phase of the effluent from UMSB reactor, while only 6% was in the effluent of UASB reactor. This amount did not include sulfide and most likely consisted mainly of elemental sulfur. Large part of the elemental sulfur oxidized during microaeration also accumulated in the G–S–L separator of UMSB reactor (10%). The accumulation of elemental sulfur in the gas space and tubes was clearly visible and identifiable as whitish solid material. This material was formed of 73% of sulfur and 20% of water. Residual 7% accounted for other elements such as P, Si, Ca, etc. The G–S–L separator of UASB reactor contained only 1% of sulfur flow with no visible accumulation in head space and it was probably present there since the start-up of the reactor.

The amount of sulfur present in the excess sludge was 8% for both UMSB and UASB reactors. No significant increase of sulfur concentration in granular sludge during the operation was observed in both reactors. I.e. elemental sulfur formed in UMSB reactor due to microaeration did not accumulated in granular sludge. The amount of sulfur bound in organic and inorganic form inside the sludge (Fig. 4) even slightly decreased by 0.5% and 2.0% in both reactors throughout the operation with the majority of sulfur bound in inorganic form (most likely as sulfide precipitates).

According to the sulfur balance calculated for 299 days for both reactors, microaeration caused the partial oxidation of inhibitory sulfide to harmless elemental sulfur in UMSB reactor. Nonetheless, no positive effect of this decrease of sulfide concentration was observed probably due to the fact that the effluent concentrations in both reactors were relatively low (Khanal and Huang, 2003). The effluent pH of 7.3 was observed in both reactors indicating that the same chemical species of sulfide (mainly HS^-) prevailed in both reactors.

3.3. The influence of microaeration on the methanogenic activity of granular sludge

Microaeration did not impair the quality of granular sludge in terms of its physical properties, the activity of methanogenic bacteria, biogas production and the efficiency of COD removal.

The efficiency of COD removal was 89 ± 5% and 90 ± 4% for UASB and UMSB reactor, respectively. COD balance was calculated for both UASB and UMSB reactors including four main effluent streams: COD in biogas, residual COD in the effluent, COD of excess biomass and COD present in the effluent in the form of sulfide and hydrogen sulfide. The fifth stream, COD consumption by oxygen dosing, was considered in UMSB reactor only. Of all COD dosed in the influent, 80% was detected in biogas from UASB while 79% of COD was detected in biogas from UMSB. The COD detected in the effluent from UASB and UMSB reactors accounted for 10% and 9% of COD dosed, respectively. COD necessary for the biomass growth accounted for 1% in both UASB and UMSB reactor. COD present in the form of sulfide and hydrogen sulfide was approximately 2% in both reactors. Air dosed into UMSB reactor accounted for 1% loss of COD. There are 7% and 8% gaps in the COD balance in UASB and UMSB reactor, respectively.

The production of biogas was 8.5 ± 2.0 and 9.6 ± 1.6 L d^{-1} for UASB and UMSB reactor, respectively (Fig. 5). The higher production of biogas in UMSB reactor (8% increase) was caused by the nitrogen dosed with air (0.8 L d^{-1}) which was inert to any processes taking place in UASB and therefore escaped in biogas. Nitrogen dilution is a disadvantage of using air instead of pure oxygen for microaeration. On the other hand, by using pure oxygen, one should considered the increased possibility of forming a dangerous and explosive gas mixture of oxygen with methane (Appels et al., 2008; Wase and Forster, 1984). However, this was not the case of this study since no oxygen was detected in the biogas. The decrease of biogas production in UASB reactor during days 280 and 320 was caused by a technical problem with the pump for substrate (smaller amount of substrate was pumped into UASB reactor).

Biogas from UMSB and UASB contained 75% and 77% of methane (after subtracting nitrogen) with the average methane production of 6.6 and 6.5 L d^{-1}, respectively. The SMA of the sludge from both reactors was measured in batch experiments performed in regular intervals (Fig. 6) and no significant difference was observed. The SMA of UMSB and UASB reactors were 0.389 and 0.336 mL CH_4 g^{-1} TSS d^{-1}, respectively, and did not significantly change during the operation. Targeted SMA experiments did not indicate trace metal limitation in either reactor (data not shown). This showed that even the high amount of sulfide formed in the reactors did not precipitate trace metals rendering them biologically unavailable.

TSS of granular sludge was 49.8 and 50.2 g L^{-1} for UMSB and UASB reactors, respectively, and did not changed significantly since the beginning of this study as the newly grown biomass was removed in regular intervals. Granules from both reactors remained compact and unchanged throughout the operation. Shen and Guiot (1996) reported that methanogens in granular sludge were more tolerant to the presence of oxygen than methanogens in flocculent sludge. Based on the multilayer structure of anaerobic granular sludge proposed by these authors, facultative anaerobes were predominant in the periphery of the granules, while oxygen-sensitive methanogens were located in the deeper

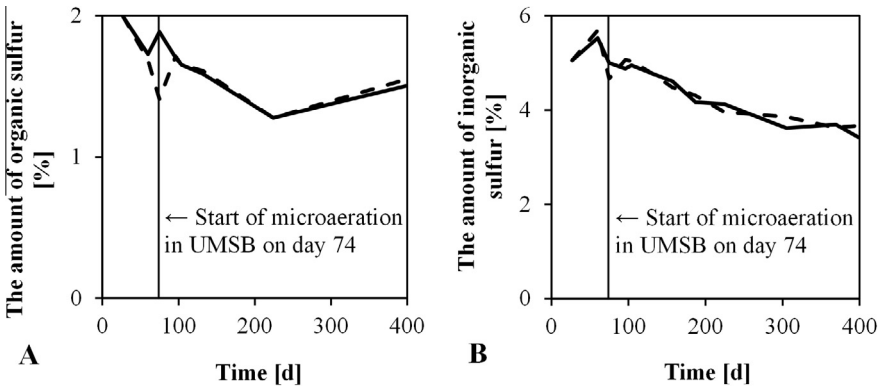


Fig. 4. Mass percentage of organic (A) and inorganic (B) sulfur in granular sludge in UMSB (---) and UASB (—) reactor throughout the operation.

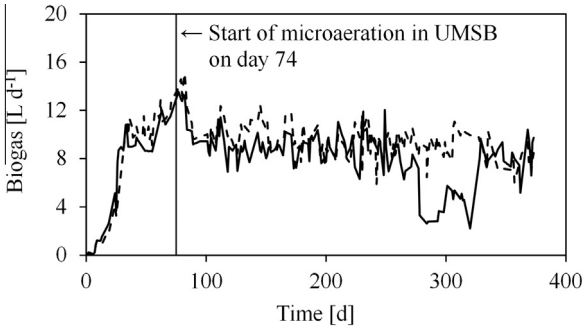


Fig. 5. The production of biogas in UMSB (---) and UASB (—) reactor throughout the operation.

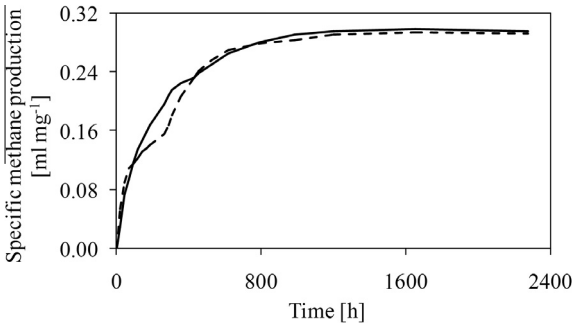


Fig. 6. Specific methane production of UMSB (---) and UASB (—) reactor after 3 months of microaerobic conditions in UMSB measured in a batch experiment.

layers, protected from the air. This explained the fact, that anaerobic organic degradation and production of methane was not affected by the dosing of air. Granules from UMSB also remained with no visible structure change. The color of the granules remained unchanged despite the findings of Zitomer and Shrout (1998), who observed the changing of color in the aerated culture from black to brown, indicating less reduction of sulfate and/or iron to ferrous sulfide.

3.4. Additional advantages of microaeration

Desulphurization of biogas is not the only advantage of microaeration. Elemental sulfur from G–L–S separator can be recovered as potential valuable compound. It is suitable for the production of sulfuric acid or it can be applied in bioleaching processes (Tichý et al., 1994). However, it is necessary to find a way how to collect elemental sulfur from UMSB reactor without the inhibition of anaerobic treatment. For that, further research of sulfur collection is needed most likely in full-scale. Elemental sulfur present in the effluent of UMSB reactor in the form of suspended solids can be used for sulfur oxidizing autotrophic denitrification (Zhou et al., 2011) since biologically produced elemental sulfur is more available than chemically produced sulfur (Kleinjan et al., 2003).

Microaeration can also enhanced hydrolysis by increasing hydrolytic production of enzymes (Johansen and Bakke, 2006). Since hydrolysis is often considered as the bottleneck for the anaerobic digestion of solids (Myint et al., 2007), optimizing this step could enhanced the whole process. Dosing small amount of air can also be a good strategy to deal with accumulation of volatile fatty acids in the system (Zitomer and Shrout, 1998) or can enhance the treatment of persistent, eco-toxic or resistant organic waste which are difficult to treat either by aerobic or anaerobic treatment itself (Botheju and Bakke, 2011). This, however, was not confirmed in this study. Last but not least, oxygen in anaerobic digestion can improve the stability of the process during unbalanced conditions (Ramos and Fdz-Polanco, 2013).

4. Conclusions

- Microaeration has the potential to remove large quantities of hydrogen sulfide from biogas formed by the anaerobic treatment of sulfur rich wastewater in UMSB reactor with granular sludge.
- Sulfur removed from the biogas and liquid in UMSB reactor was present as inorganic suspended solids in the effluent and partly accumulated on the wall of head space and in G–L–S separator.
- No negative effect on methanogenic activity of granular sludge or production of biogas was observed in UMSB reactor.
- Elemental sulfur did not accumulate in the granular sludge of UMSB reactor.

Acknowledgements

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References

Abatzoglou, N., Boivin, S., 2009. A review of biogas purification processes. Biofuels, Bioprod. Bioref. 3 (1), 42–71.
American Public Health Association, 1997. Standard Methods for Examination of Water and Wastewater, 19th ed. American Public Health Association, American Water Works Association and Water Environment Federation, Washington DC, USA.
Angelidaki, I., Alves, M., Bolzonella, D., Borzacconi, L., Campos, J.L., Guwy, A.J., Kalyuzhnyi, S., Jenicek, P., van Lier, J.B., 2009. Defining the biomethane potential (BMP) of solid organic wastes and energy crops: a proposed protocol for batch assays. Water Sci. Technol. 59 (5), 927–934.
Annachhatre, A.P., Suktrakoolvait, S., 2001. Biological sulfide oxidation in a fluidized bed reactor. Environ. Technol. 22 (6), 661–672.
Appels, L., Baeyens, J., Degrève, J., Dewil, R., 2008. Principles and potential of the anaerobic digestion of waste-activated sludge. Prog. Energy Combust. Sci. 34 (6), 755–781.
Botheju, D., Bakke, R., 2011. Oxygen effects in anaerobic digestion – a review. Open Waste Manage. J. 4, 1–19.
Buisman, C.J.N., Geraats, B.G., Ijspeert, P., Lettinga, G., 1990. Optimization of sulphur production in a biotechnological sulphide-removing reactor. Biotechnol. Bioeng. 35 (1), 50–56.
Díaz, I., Fdz-Polanco, M., 2012. Robustness of the microaerobic removal of hydrogen sulfide from biogas. Water Sci. Technol. 65 (8), 1368–1374.
Díaz, I., Pérez, S.I., Ferrero, E.M., Fdz-Polanco, M., 2011. Effect of oxygen dosing point and mixing on the microaerobic removal of hydrogen sulphide in sludge digesters. Bioresour. Technol. 102 (4), 3768–3775.
Fermoso, F.G., Bartacek, J., Manzano, R., van Leeuwen, H.P., Lens, P.N.L., 2010. Dosing of anaerobic granular sludge bioreactors with cobalt: impact of cobalt retention on methanogenic activity. Bioresour. Technol. 101 (24), 9429–9437.
Hulshoff Pol, L.W., Lens, P.N.L., Stams, A.J.M., Lettinga, G., 1998. Anaerobic treatment of sulphate-rich wastewaters. Biodegradation 9 (3), 213–224.
Janssen, A.J.H., Sleyster, R., Van der Kaa, C., Jochemsen, A., Bontsema, J., Lettinga, G., 1995. Biological sulphide oxidation in a fed-batch reactor. Biotechnol. Bioeng. 47 (3), 327–333.
Jenicek, P., Keclik, F., Maca, J., Bindzar, J., 2008. Use of microaerobic conditions for the improvement of anaerobic digestion of solid wastes. Water Sci. Technol. 58, 1491–1496.
Jenicek, P., Koubova, J., Bindzar, J., Zabranska, J., 2010. Advantages of anaerobic digestion of sludge in microaerobic conditions. Water Sci. Technol. 62 (2), 427–434.
Johansen, J.E., Bakke, R., 2006. Enhancing hydrolysis with microaeration. Water Sci. Technol. 53, 43–50.
Khanal, S.K., Huang, J.C., 2003. Anaerobic treatment of high sulfate wastewater with oxygenation to control sulfide toxicity. J. Environ. Eng. 129 (12), 1104–1111.
Kleinjan, W., Keizer, A., Janssen, A.H., 2003. Biologically produced sulfur. In: Steudel, R. (Ed.), Elemental Sulfur and Sulfur-Rich Compounds I, vol. 230. Springer, Berlin Heidelberg, pp. 167–188.
Krishnakumar, B., Majumdar, S., Manilal, V.B., Haridas, A., 2005. Treatment of sulphide containing wastewater with sulphur recovery in a novel reverse fluidized loop reactor (RFLR). Water Res. 39 (4), 639–647.
Myint, M., Nirmalakhandan, N., Speece, R.E., 2007. Anaerobic fermentation of cattle manure: modeling of hydrolysis and acidogenesis. Water Res. 41 (2), 323–332.

Peng, Y., Wang, X., Wu, W., Li, J., Fan, J., 2006. Optimisation of anaerobic/anoxic/oxic process to improve performance and reduce operating costs. J. Chem. Technol. Biotechnol. 81 (8), 1391–1397.
Ramos, I., Fdz-Polanco, M., 2013. The potential of oxygen to improve the stability of anaerobic reactors during unbalanced conditions: results from a pilot-scale digester treating sewage sludge. Bioresour. Technol. 140, 80–85.
Ramos, I., Pérez, R., Fdz-Polanco, M., 2013. Microaerobic desulphurisation unit: a new biological system for the removal of H2S from biogas. Bioresour. Technol. 142, 633–640.
Sawyer, C.N., McCarty, P.L., Parkin, G.F., 2003. Chemistry for Environmental Engineering and Science. MCGRAW-HILL Higher Education.
Shen, C.F., Guiot, S.R., 1996. Long-term impact of dissolved O_2 on the activity of anaerobic granules. Biotechnol. Bioeng. 49 (6), 611–620.
Sinbuathong, N., Khaodhiar, S., Liengcharernsit, W., Sirirote, P., Watts, D., 2007. Effect of sulfate on the methanogenic activity of a bacterial culture from a brewery wastewater during glucose degradation. J. Environ. Sci. 19 (9), 1025–1027.
Syed, M., Soreanu, G., Falletta, P., Béland, M., 2006. Removal of hydrogen sulfide from gas streams using biological processes – a review. Can. Biosys. Eng. 48, 2.1–2.14.
Tarn, K., 2002. Removal of multiple substrates in a mixed culture process for the treatment of brewery wastewater. In: Department of Agricultural and Biosystems Engineering. McGill University, Montreal.
Tichý, R., Janssen, A., Grotenhuis, J.T.C., Lettinga, G., Rulkens, W.H., 1994. Possibilities for using biologically-produced sulphur for cultivation of *Thiobacilli* with respect to bioleaching processes. Bioresour. Technol. 48 (3), 221–227.
Wase, D.A.J., Forster, C.F., 1984. Biogas – fact or fantasy. Biomass 4 (2), 127–142.
Zheng, S., Cui, C., 2012. Efficient COD removal and nitrification in an upflow microaerobic sludge blanket reactor for domestic wastewater. Biotechnol. Lett. 34 (3), 471–474.
Zheng, S., Cui, C., Quan, Y., Sun, J., 2013. Microaerobic DO-induced microbial mechanisms responsible for enormous energy saving in upflow microaerobic sludge blanket reactor. Bioresour. Technol. 140, 192–198.
Zhou, W., Imai, T., Ukita, M., Li, F., Yuasa, A., 2007. Effect of limited aeration on the anaerobic treatment of evaporator condensate from a sulfite pulp mill. Chemosphere 66 (5), 924–929.
Zhou, W., Sun, Y., Wu, B., Zhang, Y., Huang, M., Miyanaga, T., Zhang, Z., 2011. Autotrophic denitrification for nitrate and nitrite removal using sulfur-limestone. J. Environ. Sci. 23 (11), 1761–1769.
Zhu, M., Lü, F., Hao, L.-P., He, P.-J., Shao, L.-M., 2009. Regulating the hydrolysis of organic wastes by micro-aeration and effluent recirculation. Waste Manage. 29 (7), 2042–2050.
Zitomer, D.H., Shrout, J.D., 1998. Feasibility and benefits of methanogenesis under oxygen-limited conditions. Waste Manage. 18 (2), 107–116.